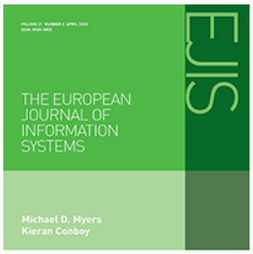

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EMPIRICAL RESEARCH

Continuous improvement of information security management: an organisational learning perspective

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ABSTRACT
This study explores ways to empower organisations to continuously improve their information security management (ISM). Drawing upon the dynamic capabilities approach, we investigated the mechanism wherein absorptive capacity has an effect. We found that absorptive capacity affects an organisation's continuous improvement of ISM, with its effect mediated through an organisation's adaptability to information security threats. In addition, the effect of absorptive capacity on adaptability is contingent upon the organisation's competitive pressure, which enhances the mediating effect of adaptability. We tested our research model using survey data collected from 130 US-based managers familiar with information security management in their organisations. Theoretical and practical implications of the study are discussed.

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Information security management; continuous improvement; dynamic capabilities; absorptive capacity; adaptability; strategic flexibility; competitive pressure

1. Introduction

As organisations increasingly rely on information and communication technologies to improve business outcomes, they have become attractive targets to cybercriminals (Jouini et al., 2014). Constantly evolving, information security threats are specifically crafted to exploit different organisations (Carr, 2010). Accordingly, organisations have strived to identify and determine ways to fight against them (Hoffman et al., 1999; Jouini et al., 2014).

Organizations often employ systematic approaches to protect their digital assets against security threats: the so-called information security management system[1] (ISMS). ISMS involves control and coordination to protect organisations' valuable assets (Darby, 2018). More specifically, it entails a set of procedures, guidelines, and policies for comprehensively identifying vulnerable information assets, establishing security objectives, evaluating risks, and taking actions to deal with those risks (Dubois et al., 2010; Stoneburner et al., 2002). For example, the ISO 27k family of standards, such as ISO 27001 and ISO 27002, provides a set of guidelines and codes of conduct to certify robust information security management (ISM) in organisations (Itgovernance, 2018).

Given that business environment and its requirements modify on a regular basis, managing information security has become a challenge for organisations and requires constant attention of management team. With the emergence of new cybersecurity threats, organisations need to continuously identify gaps, assess security performance, and improve the effectiveness and efficiency of their security practices.[2] An important element of ISM is the inclusion of continuous monitoring and improvement to meet evolving security threats (Fogalin, 2009; General, 2002) so that the organisation can retain its competitive edge and respond to the constantly changing landscape (Gonzalez & Martins, 2016). Besides dealing with ongoing concerns, continuous improvement provides organisations the opportunity to not only understand the underlining process that lead to security issues but also adjust accordingly (Gelnaw, 2019). Such efforts gradually enhance the effectiveness of ISM. Furthermore, continuous improvement facilitates the adoption of the latest changes. "Continual improvement for ISO 27001 clause 10.2", for example, discusses the mechanisms for continual evaluation and upgrades as well as the requirements and practices for ongoing improvement.

We define *continuous improvement of information security management* (CIISM) as persistent efforts to improve an ISMS (Anand et al., 2009; Peng et al., 2008). CIISM is an active ongoing effort to search for and employ measures and approaches to secure information assets (Anand et al., 2009). It aims to enhance information security effectiveness on an ongoing basis, and thereby sustain an organisation's competitive advantages (Gonzalez & Martins, 2016; Kohlbacher, 2013). Such improvement takes into account persistent actions, such as frequent monitoring of information security risks to minimise the likelihood of attacks, as well as periodical

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projects, such as a critical system upgrade to protect against new malware (Lillrank et al., 2001). Although CIISM is critically important, the primary drivers for organisations to continuously improve their security management are still unclear. Prior literature in information security has limited exploration.

Since continuous improvement activities require organisations to implement small to medium-sized changes (Smadi, 2009), we draw upon the dynamic capabilities perspective (Teece et al., 1997) as an overarching conceptual umbrella to develop our research model. We argue that to continuously improve an ISMS, organisations will first develop the capability to obtain and update their knowledge base – consisting of both security and non-security related knowledge – on an active basis and then accordingly adapt and adjust organisational activities (Weick & Quinn, 1999). In other words, organisations need to build up the capability to absorb the latest knowledge from a dynamic environment.

Prior literature in management suggests that continuous improvement is a function of organisational learning (Helfat et al., 2007; Zangwill & Kantor, 1998), i.e., "the process of improving actions through better knowledge and understanding" (FIOL & LYLES, 1985, p. 803). For this study, absorptive capacity (ACAP) is defined as the organisational ability to understand the importance of new information, incorporate it into existing information, and implement the updates through competitive actions (Cohen & Levinthal, 1990). However, the role of ACAP in the context of cybersecurity management is less studied. Given the ever-changing cybersecurity environment and the significance of gaining and sustaining competitive advantage in this area, the current study aims to bridge the research gap by examining the direct effect of ACAP on CIISM. To our best knowledge, prior literature in information security has never examined the relationship between ACAP and CIISM.

To further unearth the mechanism via which ACAP affects CIISM, by taking a dynamic capabilities perspective, we introduce an organisation's adaptability to information security threats (AIST), reflecting an organisation's strategic flexibility in security management, meditates the effect of ACAP on CIISM. We define AIST as the degree to which an organisation is capable of reconfiguring actions and activities in response to changing information security threats (Simsek, 2009; Smit, 2015). AIST allows organisations to either find new techniques or change current ones to adapt to emerging security threats (Basadur et al., 2014; Lei et al., 1996). A high level of ACAP may help organisations anticipate and analyse both potential and lesser-known information security threats (Santos-vijande et al., 2012) and subsequently, become more capable of accommodating changes in the threats. This then leads to CIISM (Boer et al., 2017; Boynton, 2007).

In addition, industry reports indicate that more than 25% of cyberattacks have a business competitive motive (Radware, 2017). In other words, cyberattacks are designed to shut down or disrupt online businesses or even steal trade secrets, thereby causing reputation and monetary losses for one's competitors (Lifars, 2020). On the other hand, competitive pressure may affect how organisations respond to threats (C-H Wang et al., 2012; Phillips, 1999). Despite the critical role of competitive pressure, there has not been enough attention or information in the literature on how organisations can use their internal capabilities to achieve higher organisational outcomes in the presence of such environmental complications. In hypercompetitive markets, organisations constantly look for ways to meet stockholder expectations and are willing to come up with and invest in exclusive tactics to enhance internal capabilities (Boone, 2000; Darai et al., 2010; Lee, 2009). In this study, competitive pressure refers to the degree of pressure an organisation feels from its competitors within the industry to implement innovative information security methods. We argue that competitive pressure could facilitate the effect of ACAP on AIST.

We, therefore, examine three research questions in this study:

(1) How does absorptive capacity influence continuous improvement of information security management in an organisation?
(2) Does organisational adaptiveness to information security threats mediate the effect of absorptive capacity on continuous improvement of information security management?
(3) Is the effect of absorptive capacity on adaptiveness to information security threats contingent upon the amount of competitive pressure?

We tested our research model using survey data collected from 130 organisations in the United States completed by managers with cybersecurity knowledge and at least three years of relevant working experience in IT-related functions. The results suggest that CIISM is both directly and indirectly influenced by ACAP, and AIST partially mediates the effect of ACAP. Furthermore, for organisations facing higher competitive pressure, the mediating effect of AIST is more salient.

2. Related literature

In this section, we first review the related literature on information security management. We then discuss the research in absorptive capacity to clarify how this construct has been theorised and applied in the information systems research.

2.1. Information security management

Information security management mainly includes guidelines for identifying vulnerable information assets, establishing security objectives, evaluating risks, and taking actions to deal with those risks (Dubois et al., 2010; Stoneburner et al., 2002). With the increasing number of cyberattacks and the extent of the associated damage, information systems researchers have paid more attention to ISM (C-W Liu et al., 2020). A few themes came to light (Hui et al., 2016), including risk management, security investments, and behavioural compliance.

The risk management literature offers tools and frameworks for managing and mitigating threats in organisations. For instance, Moeini and Rivard (2019) explained and predicted the risk response decisions of IT project managers through a risk response model. Wang et al. (2015) indicated that information systems threats differ in their risk characteristics, and such differences lead to users' distinct information-seeking behaviour. Abraham et al. (2019) discuss key contributing factors in managing cybersecurity risk and offer a roadmap to a more calculated and proactive method of managing such risks. Benz and Chatterjee (2020) proposed a tool for small and medium-sized enterprises which helps them evaluate the current level of their cybersecurity maturity and readiness.

In security investment, research mostly relies on game theory and economics analyses. For example, Safi et al. (2021) investigated the mis-spending on information security measures in organisations. They suggested that whenever a threat is perceived, decision-makers react, even if it is a minor threat. In cases of higher risk, decision-makers likely overinvest in security. With the increasing number of cybersecurity data breaches, scholars have shown interest in discovering the effect of a data breach on security investment and vice versa. For example, Kwon and Johnson (2014) suggested that investing in proactive security strategies results in fewer security failures in the healthcare sector. Wang et al. (2008) proposed a value-at-risk approach to not only measure the risk of daily losses an organisation faces due to security exploits but also estimate the value at risk.

A set of studies examined factors that encourage employees' behaviour beneficial to organisational security or discourage a detrimental one. Such studies offer managers a better understanding of employee behaviour, which helps them plan their organisation's information security strategies more precisely. For example, AW Cram et al. (2019) provided a guideline to manage information security policy (ISP) compliance by offering a consensus on the main ISP compliance antecedents in various contexts. Yazdanmehr and Wang (2021) argued that peer monitoring discourages employee ISP violations. More holistically, WA Cram et al. (2017) synthesised the current state of knowledge about organisational information security policies and outlined promising directions for future research. In addition, similar studies have investigated the adoption of specific security-related technologies. For example, Ghahramani and Wang (2017) explored the factors that affect decision-makers' intention to adopt a novel authentication process as a replacement for traditional password-based authentication systems.

Despite the significant progress of studies in this area, CIISM has been barely investigated. Prior studies have focused on identifying and investigating effective design and implementation of security management practices in organisations. While it is essential to implement good ISM at the first place, it is important to continuously enhance organisational information security practices. With a mindset of constantly improving security management to cope with evolving threats (Fogalin, 2009; General, 2002), organisations may keep their competitive edge in globalised industries (Gonzalez & Martins, 2016). Lastly, as it is noted by Kotulic and Clark (2004, p. 597), the majority of studies on ISM is "based on opinion, anecdotal evidence, or experience", calling for more theoretically-driven research.

2.2. Absorptive capacity in information systems research

Absorptive capacity has long been considered necessary for the success of organisations that face market pressures and competition (Lane et al., 2006). It has also been studied in a wide range of research in information systems, including IT business value (Bhatt & Grover, 2005), is assimilation (Cooper & Molla, 2017), and IT success (Harrington & Guimaraes, 2005). Additionally, it has been considered in terms of intention to adopt, implement, or use in enterprise systems and knowledge management domains (Gao et al., 2017).

The organisational and IS literature has looked at ACAP from two general perspectives (Roberts et al., 2012). The first takes an asset perspective in which it is theorised based on an organisation's related prior knowledge at a single point in time. In this context, knowledge is seen as a static object, and therefore one

of its proxies is the organisation's knowledge base (Tsai, 2001). For instance, Ko et al. (2005) suggested that related prior knowledge speeds up knowledge transfer among enterprise resource planning clients and consultants.

The second perspective views ACAP as either a substantive or dynamic capability (Lane et al., 2006). The substantive capability view uses organisational processes and routines to detect, absorb, transform, and employ knowledge. For instance, a software development team's expertise integration would be influenced by its ACAP, which consecutively accelerates team creativity (Tiwana & Mclean, 2005). Compensation policies or knowledge-sharing routines are examples of the ways scholars have operationalised this view (Lane et al., 2001; Roberts et al., 2012).

Second, the dynamic capability view of ACAP moves one step further and requires existing substantive capabilities for reconfiguration or modification based on the recently gained knowledge. For instance, Malhotra et al. (2005) found that engaging in inter-organisational processes and routines gives organisations the opportunity to process the obtained information from supply chain partners and consequently enhance their capability to detect, absorb, transform, and employ external knowledge. Surveys are often used to measure this form of ACAP (Pavlou & El Sawy, 2006).

While the latter perspective sounds more promising, not many studies "explicitly conceptualize absorptive capacity as a dynamic capability" (Roberts et al., 2012). A more suitable option for ever-changing environments, ACAP highlights the importance of innovation and reforming processes and resources, which is necessary to gain and sustain a competitive advantage. We, therefore, theorised it from the perspective of dynamic capabilities.

3. Theoretical background

3.1. Dynamic capabilities approach

Nowadays, organisations are in a hypercompetitive environment and face frequent changes in regulatory, social, and technological domains (Barreto, 2009). Failure to address such changes can negatively impact performance (Audia et al., 2000). Therefore, effectively coping with ongoing changes is necessary to maintain desired outcomes (Eisenhardt & Martin, 2000). The dynamic capabilities approach (Teece et al., 1997) has been increasingly used in strategic management literature to deal for this reason.

Dynamic capabilities, by definition, refer to an organisation's alignment in integrating, reconfiguring, renewing, and reconstructing its resources and capabilities continuously, as well as updating and rebuilding its primary capabilities (Wang & Ahmed, 2007). Organizations primarily invest in them as an ultimate capability that contributes to long-term outputs, such as performance (Teece et al., 1997). Dynamic capabilities operate as an underlining process, thereby transforming organisational resources and capabilities into a product, service, or process consistent with recent industry and market changes (Wang & Ahmed, 2007). In fact, they provide a source of sustained competitive advantage.

To successfully operate in dynamic and turbulent environments and retain a competitive advantage, organisations must continuously put their everyday learning into action (Zahra & George, 2002). ACAP, considered a specific form of organisational learning, reflects a dynamic capabilities perspective (Sun & Anderson, 2010; Zahra & George, 2002). It increases strategic flexibility and helps in formulating innovative activities (Clauss et al., 2021). More specifically, strategic flexibility is also a dynamic capability that emphasises resource adaptability and process reconfiguration (Eisenhardt & Martin, 2000). Building and maintaining it ensures organisations keep up with this agility and retain their competitive edge (Adner & Helfat, 2003). To build this flexibility, organisations can incrementally deepen their existing capabilities by learning and incorporating new practices (Sirmon et al., 2007).

In particular, ACAP accelerates and contributes to the renovation of an organisation's knowledge base and skills, which facilitates the flexible use of resources (Miroshnychenko et al., 2021). In other words, it develops sensitivity and awareness to a wider range of relevant and important exogenous developments in the market that might be significant to upcoming operations. This strengthens the ability to flexibly adapt and reconfigure processes and resources (Nowak, 2021). Therefore, organisations with well-developed ACAP are more likely to detect new opportunities and reconfigure their processes and resources accordingly (Zahra & George, 2002).

Moreover, strategic flexibility enables organisations to develop and enhance new managerial capabilities to help them quickly respond to or take control of emerging threats and challenges. This leads to better outcomes (Dalziel, 2009; Wang et al., 2015). Besides, ACAP provides the opportunity to distinguish the importance of applicable exogenous changes and how it can enhance current processes. For this reason, ACAP would lead to better organisational outcomes as well (Nowak, 2021). In addition, higher levels of strategic flexibility help organisations decouple from routine processes and functions more easily and smoothly adapt to the reconfigured routines with less costs (Sanchez, 1995). Thus, it can accelerate adaptation to new or reconfigured processes, and thereafter, enhance

outcomes. This indicates the possible mediating role of strategic flexibility on the effect of ACAP on organisational outcomes.

Furthermore, prior studies have highlighted the critical role of strategic flexibility in competitive environments (Grewal & Tansuhaj, 2001). It affords organisations the opportunity to manage rival moves and cope with changes in the environment (Combe et al., 2012). High competition makes it more difficult to gain and sustain a competitive advantage. The resulting pressure makes the capabilities to learn more essential. This suggests the moderating role of competitive pressure on the effect of ACAP on strategic flexibility. Figure 1 depicts the proposed conceptual model.

3.2. Absorptive capacity as a dynamic learning capability

From a practical perspective, organisations must understand what they expect to happen in the future is not simply a not-yet-happened extension of the past. Instead of conditioning the future based on past projections, organisations need to know that the "future" may arrive, which requires taking actions as time goes on. Focusing on "knowing" and learning as a dynamic capability can originate such understanding and empower organisations to continuously find out about the latest information and come up with real-time creative responses and improvisation (Purser & Petranker, 2005). Accordingly, we explore the mechanism through which organisational learning affects continuous improvement in information security management (as an organisational outcome).

More specifically, ACAP refers to organisational learning from a dynamic capability perspective (Sun & Anderson, 2010; Zahra & George, 2002). It is defined as an organisation's ability to realise the value of new external knowledge and thereafter assimilate and apply it to competitive actions and innovations (Cohen & Levinthal, 1990; Roberts et al., 2012; Tu et al., 2006; Zahra & George, 2002). ACAP enables quick examination and interpretation of environmental and organisational changes. Using the interpreted information, organisations would be able to originate essential reconfiguration, rearrangement, and renewal of associated capabilities, thereby enhancing their flexibility in response to various technological, competitive, and demand uncertainties (Patel et al., 2012).

Following the established literature, organisational ACAP is captured by four aspects: acquisition, assimilation, transformation, and exploitation (Jansen et al., 2005; Malhotra et al., 2005; Zahra & George, 2002). *Acquisition* emphasises the ability to recognise, gather, and develop new and relevant knowledge from various sources (Liu et al., 2013; Setia & Patel, 2013). Quality of acquisition is mainly determined by the amount and speed of the effort to recognise, collect, and import such knowledge (Kim, 1997). Next, *assimilation* focuses on the ability to absorb and understand the newly gained knowledge (Liu et al., 2013; Setia & Patel, 2013). This is assessed by the extent to which an organisation establishes the routines and processes that allow effectively analysing and interpretation of the knowledge gained from external sources (Kim, 1997; Szulanski, 1996). Third, *transformation* reflects the ability to integrate existing and newly gained knowledge (Liu et al., 2013; Setia & Patel, 2013). This can be achieved either by modifying (adding or deleting) the existing knowledge or interpreting it in a different way (Zahra & George, 2002). Lastly, *exploitation* describes the ability to apply new knowledge to accomplish objectives and meet market requirements (Liu et al., 2013; Setia & Patel, 2013). The outcome is the constant creation of innovative processes, products, and systems (Spender, 1996).

While acquisition and assimilation (the so-called *potential* subset of ACAP) involve recognising and absorbing external knowledge, transformation and exploitation (the so-called *realised* subset of ACAP) entail leveraging the absorbed knowledge (Lane & Lubatkin, 1998; Zahra & George, 2002). Each subset is required but not sufficient on its own to enhance performance.

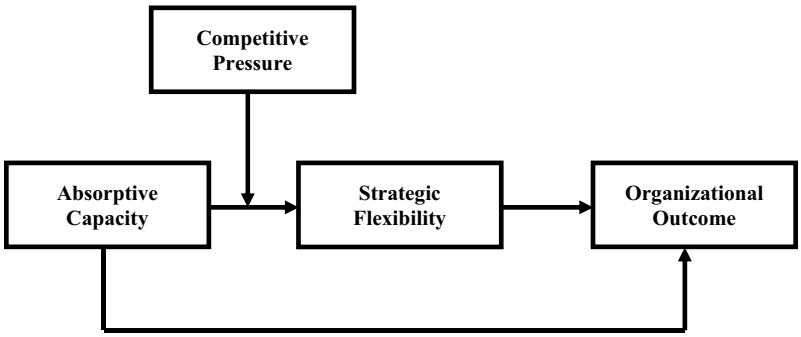

Figure 1. Conceptual model.

ACAP can be considered an essential organisational learning capability that enables knowledge management (Malhotra et al., 2005; Wheeler, 2002; Zahra & George, 2002). It helps explain why some organisations are more capable of effectively managing environmental uncertainties (Patel et al., 2012). In fact, ACAP consists of a series of actions to manage a company's existing knowledge and build new knowledge based on prior related knowledge, rich communication, and learning routines (Malhotra et al., 2005; Tu et al., 2006). An organisation exhibiting a high level of ACAP is more adept at processing new information and utilising skills and knowledge (Lane et al., 2006; Zahra & George, 2002). As a result, such an organisation is more capable of taking effective and efficient actions in response to new challenges (Cohen & Levinthal, 1989).

Additionally, organisations with a high level of ACAP are good at applying newly obtained knowledge in various domains (Roberts et al., 2012). This knowledge may not be necessarily limited to information security in a straight line but can contribute to the information security domain to some extent. For instance, it can complement and improve an existing knowledge base (Van Den Bosch et al., 1999) in two ways: (1) enrich information security knowledge, such as details about recently identified vulnerabilities and new types of malware (Esteves et al., 2017), and (2) improve knowledge in other related domains, such as information on how to use new software to analyse data more rigorously and generate meaningful reports to support decision-making. Second, the newly gained knowledge can help with predicting technological trends (Cohen & Levinthal, 1994), such as innovations competitors may develop, or reorganising the existing capabilities and behaviours (Pavlou & El Sawy, 2006), such as effectively enhancing cybersecurity, ethics, and regulatory compliance training (Esteves et al., 2017). Finally, it may be used to create inventive services and products (Roberts et al., 2012), such as a new intrusion detection or prevention system.

3.3. Adaptability to information security threats as strategic flexibility

In the context of information security, we conceptualise strategic flexibility as adaptability to information security threats and define it as the degree to which an organisation is capable of reconfiguring actions and activities in response to changing information security threats (Golden & Powell, 2000). Such adaptability enables organisations to respond to security threats in a way to minimise the disruption of operations (Combe et al., 2012).

For adaptability, managers need to strengthen their organisation's learning capability and create a learning culture (Cepeda & Vera, 2007). By highlighting the importance of capturing and deploying the knowledge base efficiently, ACAP facilitates adaptation in high-velocity environments (Floyd & Lane, 2000). It enables organisations to capture the relevant information better, anticipate market changes, and abandon activities that are no longer effective (Santos-Vijande et al., 2012). In addition, an organisation with a high level of ACAP invests more in research and development (Tsai, 2001). This allows exposure to a wide range of knowledge, ideas, and experiences and encourages thinking outside the box (Levinthal & March, 1993a; Vasudeva & Anand, 2011).

Besides, according to dynamic capability theory (Dixon et al., 2014; Levinthal & March, 1993b; March, 1991), two categories of organisational capabilities are necessary to be adaptive over time: (1) capabilities that enable the organisation to use its existing resources (e.g., skills, processes, and knowledge) effectively and efficiently and (2) capabilities that enable the organisation to renew its competences to catch up with the changes in the environment by creating new skills, processes and knowledge via innovation, discovery, research, or test and trial. Achieving high adaptability in security management requires both efficacy and innovation along with the ability to orchestrate and integrate both into the organisation (O'Reilly & Tushman, 2008; Uhl-Bien & Arena, 2018). The aforementioned subsets of ACAP (i.e., potential and realised ACAPs) enable organisations to not only develop the required efficacy and innovation but also smooth the adaptive process.

3.4. Competitive pressure

Competitive pressure refers to the degree of pressure an organisation feels from its competitors to implement innovative methods. Prior literature has argued that even in environments in which change is slower, dynamic capabilities exist and are utilised (Zollo & Winter, 2002). They may be especially important for organisations in an environment wherein changes take place frequently (Eisenhardt & Martin, 2000).

As the business environment becomes more competitive, organisations increasingly realise the importance of implementing strategies that distinguish their services and products. In doing so, they highlight their

Table 1. Summarises the definitions of the key constructs in this research.

Construct	Definition
Continuous Improvement in Information Security Management (CIISM)	An active ongoing effort to search for and employ measures and approaches to secure information assets in an organisation.
Absorptive Capacity (ACAP)	The ability of an organisation to understand the importance of new information, incorporate it into existing knowledge, and employ the updated version in competitive actions.
Adaptability to Information Security Threats (AIST)	The degree to which an organisation is capable of reconfiguring actions and activities in response to changing information security threats.
Competitive Pressure	The degree of pressure an organisation feels from its competitors to implement innovative information security methods.

unique advantages to prospective customers and other important market participants (Hsu et al., 2012) in altering the competitive landscape (Porter & Millar, 1985). Therefore, competitive pressure pushes organisations to establish and sustain competitive advantages over their competitors (Zhu & Kraemer, 2005). Table 1 summarizes the definitions of the key constructs in this research.

4. Hypothesis development

In this section, we propose that absorptive capacity enhances an organisation's continuous improvement of information security management through an organisation's adaptability to information security threats. Additionally, the impact of absorptive capacity on adaptability is more pronounced when the organisation faces greater competitive pressure. Finally, competitive pressure enhances the mediating effect of adaptability. The proposed research model is illustrated in Figure 2.

As continuous improvement is mainly determined by the capabilities to obtain and process new information (Helfat et al., 2007; Zangwill & Kantor, 1998), organisations with higher ACAP are better at improving their information security practices in a sustainable way by acquiring and using timely and useful information

Table 2. Key Demographic Variables, Title, and Work Experience of Respondents (N = 130).

Variable		No.	Percent	Variable		No.	Percent
Age	25 to 34 years	36	27.5%	Race	White/Caucasian	96	73.8%
	35 to 44 years	30	23.1%		African American	7	5.4%
	45 to 54 years	34	26.2%		Hispanic	7	5.4%
	55 to 64 years	28	21.5%		Asian	17	13.1%
	65 years and over	2	1.5%		Native American	2	1.5%
					Pacific Islander	1	0.8%
Education	High School/GED	4	3.1%	Title	President, Owner, or Managing Director	8	6.2%
	Some College/Bachelor	58	44.6%		CIO/CTO/VP of IS	23	17.7%
	Master's degree	59	45.4%		IS Manager, Director, Planner	60	46.2%
	Professional degree	6	4.6%		Other Manager in IS department	35	26.9%
	Doctoral degree	3	2.3%		Business Operations Manager	2	1.5%
					Administration/Finance Manager	2	1.5%
Income	20 K to 40 K	4	3.1%	Work Experience	Less than a year	7	5.4%
	40 K to 60 K	19	14.6%		1 to 3 years	12	9.2%
	60 K to 80 K	28	21.5%		3 to 5 years	23	17.7%
	More than 80 K	79	60.8%		5 to 10 years	45	34.6%
					More than 10 years	43	33.1%
Gender	Male	89	68.5%				
	Female	41	31.5%				

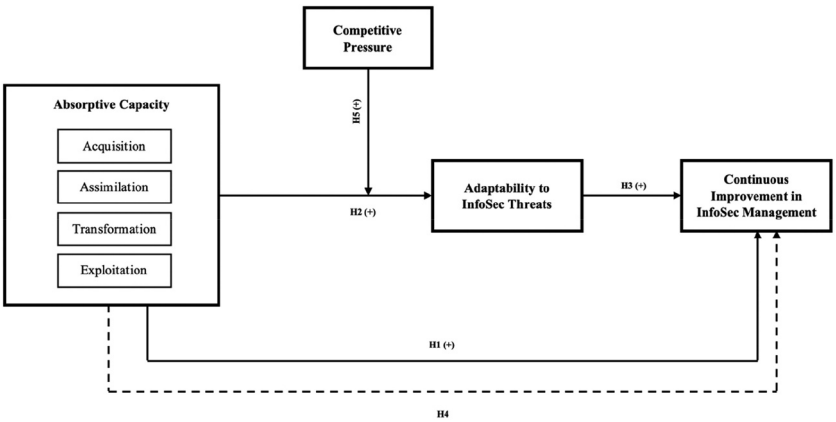

Figure 2. Research model.

(Anand et al., 2009; Linderman et al., 2004). Such information helps organisations be aware of the latest prevention, detection, and responsive actions in the industry and gradually use them to enrich various aspects of information security management.

Moreover, organisations used to obtaining and applying the latest available knowledge have a more accurate understanding of why continuous improvement in information security is important, what types of improvement practices are more effective, and how to perform them faster and more effectively (Anand et al., 2009; Zangwill & Kantor, 1998). With this in mind, organisations with higher ACAP are better able to use their knowledge base to create ongoing improvements to enhance information security management. Even if the majority of these improvement programs include minor steps and may have little impact, they will make a considerable contribution to the organisation's security management over time (Bessant et al., 1994), which is the main objective of CIISM. Thus, we hypothesise:

H1. *An organisation's absorptive capacity is positively related to its continuous improvement in information security management systems.*

Further, we argue that organisations with a higher level of ACAP are more proficient in acquiring applicable information and persistently updating their knowledge base. The updates enhance adaptability in terms of recognising problems and opportunities extant in their environment (Beer et al., 2005; Lumpkin & Lichtenstein, 2005). This leads to a higher capability of anticipating and discovering unknown and less-known security threats and vulnerabilities while finding appropriate and innovative means to respond to them (Iyengar et al., 2015; Malhotra et al., 2005; Zahra & George, 2002). Organizations are then more prepared to rapidly allocate resources to discover new threats and ways to neutralise them (Argyris & Schon, 1978; Santos-Vijande et al., 2012). Accordingly, we hypothesise:

H2. *An organisation's absorptive capacity is positively related to its adaptability to information security threats.*

Contemporary firms must handle more sophisticated and targeted attacks and thus, need to constantly improve security management processes. Examples of such attacks include phishing or ransomware, which can remain undetected for an extended period. Adaptability to information security threats empowers organisations to neutralise or lessen any harm (Simsek, 2009; Smit, 2015). AIST becomes more important when firms have to deal with dynamic, less known, and customised threats as there is often no predefined plan or timeline to handle and cope with them. For instance, when a spear-phishing email is designed to specifically target a financial manager of a small business with commercial banking relationships (Merkow & Breithaupt, 2014), existing measures against historically common threats will not be effective as they are meant to handle well-known or expected threats (Baskerville et al., 2014). In this scenario, AIST would enable this small business to timely respond to the unique aspects of this threat and assign resources to deal with it.

AIST prepares organisations to cope with a wide range of emerging security threats and thereby adds to the efficacy of security management in general. It follows that a higher level of adaptability facilitates incorporating ongoing security management changes (Boer et al., 2017; Boynton, 2007) and improving such processes. We hypothesise:

H3. *An organisation's adaptability to information security threats is positively related to its continuous improvement in information security management systems.*

Based on our three hypotheses (H1- H3), an organisation's ACAP could influence its capacity to continuously improve its overall ISMS by fostering AIST. Organizations with high ACAP constantly build up their information security knowledge base, which enables them to be more informed and prepared for potential and less-known information security threats (Santos-Vijande et al., 2012). In a dynamic and ever-changing context, such as the one with increasing challenges to information security, ACAP is a necessity to obtain competitive advantage. However, to remain competitive and sustain the obtained competitive advantage, flexibility and ability to adapt processes to the latest security threats are important as well. Such adaptability enables organisations to keep up with ongoing changes (Adner & Helfat, 2003).

ACAP also empowers organisations to be more cautious about routine variations in security practices and to allocate the required resources to deal with less-known security threats more effectively and quickly. This adaptation will accelerate the process of discovering innovative methods to handle security issues and smooth out the process of constant improvement to organisational ISMS. So, we hypothesise:

H4. *An organisation's adaptability to information security threats mediates the relationship between its absorptive capacity and its continuous improvement in information security management systems.*

High level of competitive pressure may make organisational capability more valuable (Aragón-Correa & Sharma, 2003). Organizations are aware of the

importance of responding to competitive pressure to enhance their standing among rivals (Dai et al., 2018; Freeman, 2010), and many are willing to develop a unique approach to fulfill stakeholder expectations (Liu et al., 2010). An example is adopting a new technology or implementing a novel technique less used by competitors.

Another consideration is that elevated competitive pressure may lead to a rise in illegal access to information and novel attacks (Merkow & Breithaupt, 2014). Therefore, in a market with higher competition, organisations may more likely have to rely on internal capabilities and resources, such as absorptive capacity, to create efficient and innovative adaptive measures to cope with security threats (Aragón-Correa & Sharma, 2003) in order to enhance the chance of survival (Bansal & Roth, 2000; Deephouse, 1996; Delmas & Toffel, 2008).

Low level of competitive pressure, however, may not create a sense of urgency for organisations to explore different ways of detecting potential information security threats on a routine basis. In such cases, organisations may not feel the need to invest in and rely on their capabilities that can help them adapt to security threats. On the other hand, the sense of urgency in a market with higher competition can also help organisations gain a competitive advantage that will enable them to thrive and grow (Colwell & Joshi, 2013), which indicates a stronger link between ACAP and AIST. Therefore, we hypothesise:

H5. *The effect of absorptive capacity to adaptability to information security threats is stronger for organisations that face more competitive pressure.*

Collectively, we have proposed that (a) ACAP positively influences AIST (H2), (b) AIST affects CIISM (H3), (c) AIST mediates the relationship between ACAP and CIISM (H4), and (d) Competitive pressure moderates the relationship between ACAP and AIST (H5). Conceptually, our hypotheses combined a mediation effect (i.e., H4, the impact of ACAP on CIIST through AIST) and a moderation effect of competitive pressure on the path between ACAP and AIST (H5). This structure is consistent with the configurations of a moderated mediation model (Edwards & Lambert, 2007; Muller et al., 2005). In particular, it reflects a first-stage moderated mediation model, indicating that the magnitude of the mediating process is contingent on the level of the moderator (Preacher et al., 2007).

To describe the combined effects of moderation and mediation, each effect could be analysed separately by testing the significance of their corresponding path, and results from these analyses are interpreted together. However, research has shown that such an approach has several conceptual limitations (see, Edwards and Lambert (2007) for further discussion). Thus, it is recommended for researchers to propose the moderated-mediated effect and test such an effect using an analytical framework that incorporates moderated mediation (Edwards & Lambert, 2007).

Empirically speaking, assuming that competitive pressure moderates the association between ACAP and AIST, it is also likely that the strength of the indirect relationship between the organisation's ACAP and its CIISM via AIST will be affected by competitive pressure – thereby demonstrating a pattern of moderated mediation between the studied variables. In light of our expectation that there is a strong (weak) indirect relationship between the organisation's ACAP and its CIISM when competitive pressure is high (low), we propose the following hypothesis:

H6. *The mediating effect of AIST is stronger for organisations that face more competitive pressure.*

5. Research methodology

5.1. Sample and data collection

We employed a field survey method to collect data for hypothesis testing using Qualtrics Panel Services. To ensure data quality, we followed the latest methodological literature on online survey platforms (LOWRY et al., 2016; STEELMAN et al., 2014). Additionally, three attention check questions were used to filter out respondents who did not pay attention. The focus of our study is continuous improvement, wherein managers often handle its design, planning, operation, and monitoring. Thus, given their responsibility in implementing and overseeing continuous improvement practices, managers are the best source to gather insight into such practices. We, therefore, restricted our survey respondents to managers with technical knowledge about the security management in their organisations. We further refined our sample to only include those with more than three years of working experience in IT-related functions. As a result, the sample size was reduced to 130 from 193. Close to 70% of the respondents were male, and about 77% were within the age range of 25 to 54. Furthermore, nearly 68% had more than five years of work experience. Nearly 45% were from organisations with more than 1,000 members, while more than 50% were from organisations that had existed for more than 26 years. Tables 2 and 3 summarise this information.

5.2. Measurement items

To ensure the reliability and validity of the measurement items, we adapted existing instruments from prior studies. All the constructs were modelled as reflective and measured using a five-point Likert scale. The CIISM

Table 3. Industry, Size and Age of Organizations.

Variable		No.	Percent
Organization Size	Less than 100 employees	16	12.30%
	100–400 employees	28	21.50%
	401–700 employees	12	9.20%
	701–1000 employees	16	12.30%
	More than 1000 employees	58	44.60%
Organization Age	Less than 5 years	2	1.50%
	5 to 10 years	14	10.80%
	11 to 15 years	18	13.80%
	16 to 25 years	26	20.00%
	More than 26 years	70	53.80%
Industry	Education	9	6.90%
	Finance	15	11.50%
	Wholesale/Retail	14	10.80%
	Healthcare	13	10.00%
	Construction	2	1.50%
	Insurance	2	1.50%
	Other	75	57.70%

measurement items were adapted from HYATT and RUDDY (1997) and reworded to correspond to the context of our study. Second, ACAP measurement items (covering the aspects of acquisition, assimilation, transformation, and exploitation) were adapted from PAVLOU and El Sawy (2006). Next, AIST measurement items were adapted from De JONG and De Ruyter (2004) and SPIRO and WEITZ (1990). Lastly, measurement items for competitive pressure were adapted from LIN (2006) and GRANDON and PEARSON (2004).

The initial questionnaire was reviewed by two information security management experts to ensure face validity. The survey was then revised based on their comments and feedback. A pilot study was conducted among 32 master's students of MIS at a large university in the southwestern region of the United States. Appropriate changes to the survey were made based on the results. The support based on both the literature and the pilot study allowed us to confirm the content validity of our instrument and measures (STRAUB et al., 2004). The final measurement items are provided in Table A1 of the Appendix.

6. Data analysis and results

6.1. Method of analysis

The analysis for this study was conducted in two stages. First, the measurement model was evaluated using a confirmatory factor analysis (CFA) in IBM AMOS v24 software. Second, the research model was tested using the ordinary least squares (OLS) regression analysis in IBM SPSS v23 software. We followed this two-stage approach because it has been not only stated to be the most appropriate for model estimations involving interaction effects but also used in recent studies (e.g., MAGNI et al., 2018; MARUPING & MAGNI, 2012; MARUPING et al., 2019; RAI & TANG, 2010; TANRIVERDI & UYSAL, 2011; WINDELER et al., 2017). In addition, we chose OLS over the structural equation model (SEM) technique to test the research hypotheses because SEM may not be robust for relatively small sample sizes, especially for models with interaction terms (MARUPING et al., 2019).

6.2. Measurement validity

To assess measurement model fit, we followed HU and BENTLER's (1999) recommended fit indices, including chi-square (χ^2) with degree of freedom (DF), non-normed fit index (NNFI), comparative fit index (CFI), standardised root mean square residual (SRMR), and root mean square error of approximation (RMSEA). Table 4 shows these indexes and their thresholds.

To ensure measurement quality for all the constructs, the latent constructs' psychometric properties, including convergent validity, individual item reliability, composite reliability, and discriminant validity, were evaluated. As explained earlier, all the latent constructs were reflectively modelled. For each latent construct, path loadings were calculated. We observed, at Alpha level of 0.05, all the measurement items' path loadings are significant, with a value of more than 0.7 (CHIN & MARCOLIN, 1995), which indicates that more than 50% of the variance is shared between each construct's items (CHIN, 1998; see, Table A2 in Appendix).

To examine the convergent validity, composite reliability and average variance extracted (AVE) were obtained (HAIR et al., 1995). All of the constructs met the acceptable score of 0.5 for AVE, thereby confirming their reliability (FORNELL & LARCKER, 1981). Moreover, the values of composite reliability and Cronbach's alpha are above the cut-off value of 0.7 (FORNELL & LARCKER, 1981; NUNNALLY & BERNSTEIN, 1994). The square root of AVE was also used to verify the discriminant validity (FORNELL & LARCKER, 1981). For each latent construct, the square root of the AVE is higher than all its cross-correlations. Moreover, higher values of inter-construct correlations confirm the greater variance among each construct's specific measures compared to other measures (GEFEN et al., 2000). Collectively, all the indicators in the measurement model are valid through their constructs. The results are shown in Table 5.

Table 4. Measurement Model Fit Index.

Measure	Threshold	Measurement Model
χ^2	–	233.388
DF	–	162.000
χ^2/DF	Between 1 and 3	1.441
NNFI	>0.95	0.957
CFI	>0.95	0.961
SRMR	<0.08	0.052
RMSEA	<0.06	0.058

Table 5. Reliability, Average Variance Extracted and Construct Correlation Matrix.

	Means (S.D.)	CR	Cronbach's Alpha	AVE*	1	2	3	4
1. Absorptive Capacity	4.01 (0.70)	0.94	0.94	0.63	**0.79**			
2. AIST	3.61 (0.77)	0.81	0.77	0.59	0.74+	**0.77**		
3. CIISM	3.69 (0.90)	0.90	0.90	0.76	0.69	0.75	**0.87**	
4. Competitive Pressure	3.37 (0.96)	0.87	0.65	0.63	0.45	0.61	0.67	**0.79**

* The diagonal is the values of the square root of AVE.
+All the correlations are significant at p < .05

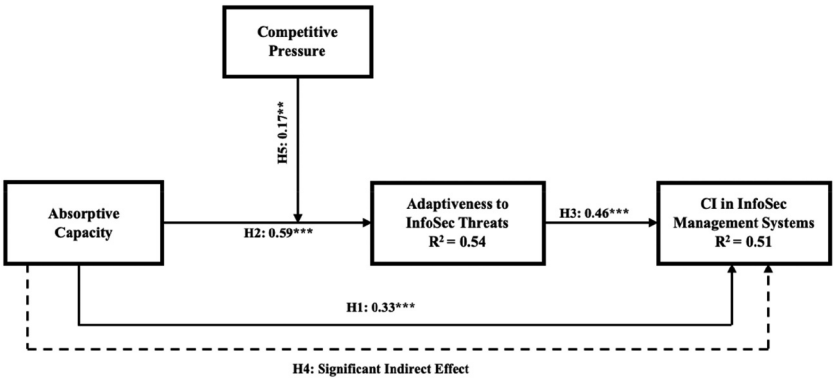

Figure 3. OLS Results.

6.3. Hypothesis testing

To run the hypothesis testing analyses, the latent factor scores for each of the constructs were calculated using SPSS. Then, two ordinary least squares regressions were performed, one with AIST and the other with CIISM as a dependent variable. The results of the data analyses, including R-squares, standardised path, and path significance, are depicted in Figure 3 and Table 6. The R-square value of CIISM and AIST are 0.51 and 0.54, respectively. The results suggest that the research model can explain nearly 54% of the CIISM. Because the amount of variances explained is larger than 10 percent, the proposed model is valid and acceptable (FALK & MILLER, 1992).

The results of testing direct relationships indicate that both ACAP (β = 0.33, p < .001) and AIST (β = .46, p < .001) have a significant positive effect on CIISM, supporting H1 and H3, respectively. The results also indicate that ACAP has a positive effect on AIST (β = .59, p < .001), as anticipated in H2. To test the mediation effects proposed in H4, we followed the Baron and Kenny method in HAYES (2009) and evaluated three paths: (a) the direct path between ACAP and CIISM (H1), (b) the path between ACAP and AIST (H2), and (c) the path from AIST to CIISM (H3). Considering that 1) both the paths in H1 and H2 were statistically significant; 2) the direct path between ACAP and CIISM (excluding AIST from the model) was significant (b = .63. p < .001); and 3) the path between ACAP and CIISM including the AIST effect (H3) was significant, we conclude that AIST partially mediates the relationship between ACAP and CIISM (HAYES, 2009; MACKINNON, 2008; SHROUT & BOLGER, 2002), thereby supporting H4. To further test the mediation effect, the indirect effect was calculated using the written SPSS macro by Preacher and Hayes (HAYES, 2013; PREACHER et al., 2007). The result suggests that the indirect effect of AIST is significant at p < 0.05, validating the support for H4 (see, Table 7).

Next, to test the moderation effect, we included the interaction between ACAP and competitive pressure. The interaction effect was calculated by multiplying the standardised values for the two variables. The path between the interaction and AIST was significant (b = .18, p < .01), supporting H5. We further validated the strength of the moderating effects by calculating

Table 6. OLS Regression Analysis.

	DV: AIST		
Predictors			
ACAP	β = .66***	β = .53***	β = .59***; VIF = 1.31
Competitive Pressure		β = .32***	β = .35***; VIF = 1.24
ACAP x Competitive Pressure			β = .18**; VIF = 1.18 *
R^2	.44	.52	.54
DV: CIISM			
Predictors			
ACAP		β = .33***	
AIST		β = .46***	
Organization Size		β = .13 n.s.	
Organization Age		β = .03 n.s.	
R^2		.51	

* p < .05; ** p < .01; *** p < .001; n.s.: not significant.

Table 7. Indirect Effect of ACAP on CIISM.

	Effect	BootSE	BootLLCI	BootULCI
AIST	0.2969	0.0600	0.1881	0.4236

the change in R^2 using Cohen's f^2, which compares the R^2 value of the interaction effect on the main effect using the following equation (CHIN et al., 2003): $f^2 = (R^2{}_{Interaction} - R^2{}_{Main})/\ (1 - R^2{}_{Main})$, yielding (0.54–0.52)/(1–0.52) ~ 0.05, suggesting that the value for H5 indicates a small to medium effect (COHEN, 1988). Then, we tested the significance of the effect by calculating the significance of the difference in variance explained between the model that contains the interaction model and the main effects model, using the following F-statistics:

$$F(df_{Interaction} - df_{main}, N - df_{Interaction} - 1) = \frac{\Delta R^2/(df_{Interaction} - df_{main})}{(1 - R^2{}_{Interaction})/(N - df_{Interaction} - 1)}$$

The F-value was 6.57 ($p < 0.001$), indicating the change in R^2 was significant. Overall, these results provide further support for H5. A small to medium effect size does not necessarily indicate an insignificant effect, especially when there is a noticeable increase in beta due to the moderations (CHIN et al., 2003). In addition, we calculated the VIF values to make sure that the interaction results do not suffer from multicollinearity (see, Table A4 in Appendix). Accordingly, we conclude that competitive pressure had a significant and substantial moderating effect on the relationship between ACAP and AIST.

Hypothesis 6 indicates that the magnitude of mediation effects proposed in H5 depends on different levels of the moderator – competitive pressure (H6). We tested the moderated mediation using the SPSS macro developed by Preacher and Hayes (HAYES, 2013; PREACHER et al., 2007). The results of this test were consistent with the results of the previous analysis. This indicates that the mediating effect of AIST differs for different levels of competitive pressure (see, Table 8–10).

To elaborate, the results in Table 8 show that the effect of ACAP on AIST is significant at all three levels of competitive pressure and this effect becomes more pronounced (from 0.39 to 071) as competitive pressure increases. In addition, the results in Table 9 show that the mediating effect of AIST is significant at all three levels of competitive pressure (there is no zero between the lower and upper confidence intervals of the mediating effects) and it is more pronounced (from 0.18 to 032) as competitive pressure increases. Moreover, the results in Table 10 show that the indirect effect of AIST is conditional on the level of competitive pressure. In other words, competitive pressure significantly moderates the indirect effect of AIST on CIISM.

Finally, to account for organisational differences in the proposed model, we followed the related literature and controlled the effect of size and age of the respondents' organisations (HSU et al., 2012). None of these variables had a marginal effect on AIST and CIISM. Table 11 summarises the hypotheses testing results.

To ensure that the common method bias (CMV) is not a concern, we adopted various procedural and statistical remedies to avoid common method biases. The CMV tests found no significant evidence of such biases (see, Table A3 in Appendix).

7. Discussion

The purpose of this study is to explore factors influencing the continuous improvement process in organisations in the context of information security management. Drawing upon the perspective of dynamic capabilities, we proposed a conceptual model wherein ACAP affects the CIISM both directly and indirectly. Our empirical findings suggest that in addition to a direct impact, ACAP indirectly influences continuous improvement through an organisation's AIST. Additionally, our findings indicate that for organisations facing higher competitive pressure, the mediating effect of AIST is stronger.

Table 8. Conditional Effects of ACAP on AIST at Different Values of Competitive Pressure*.

Competitive Pressure	Effect	se	t	p-value	LLCI	ULCI
−1.38 (Low)	0.39	0.0832	4.6386	0.000	0.2212	0.5505
0.15 (Medium)	0.60	0.0696	8.6226	0.000	0.4621	0.7374
0.92 (High)	0.71	0.0925	7.6555	0.000	0.5249	0.891

*Level of confidence for all confidence intervals in the output: 95.00; Number of bootstrap samples for percentile bootstrap confidence intervals: 5000

Table 9. Mediating Effect of AIST (ACAP->AIST->CIISM) at Different Values of Competitive Pressure*.

Competitive Pressure	Effect	BootSE	BootLLCI	BootULCI
−1.38 (Low)	0.18	0.0636	0.0723	0.3207
0.15 (Medium)	0.27	0.0485	0.1822	0.3732
0.92 (High)	0.32	0.0542	0.2181	0.4307

*Level of confidence for all confidence intervals in the output: 95.00; Number of bootstrap samples for percentile bootstrap confidence intervals: 5000

Table 10. Index of Moderated Mediation*.

	Index	BootSE	BootLLCI	BootULCI
Competitive Pressure	0.0641	0.0285	0.0029	0.1154

*Level of confidence for all confidence intervals in the output: 95.00; Number of bootstrap samples for percentile bootstrap confidence intervals: 5000

Table 11. Summary of Results.

Hypothesis	Result
H1. An organisation's absorptive capacity is positively related to its continuous improvement in information security management systems.	Supported
H2. An organisation's absorptive capacity is positively related to its adaptability to information security threats.	Supported
H3. An organisation's adaptability to information security threats is positively related to its continuous improvement in information security management systems.	Supported
H4. An organisation's adaptability to information security threats mediates the relationship between its absorptive capacity and its continuous improvement in information security management systems.	Supported
H5. The effect of absorptive capacity to adaptability to information security threats is stronger for organisations that face more competitive pressure.	Supported
H6. The mediating effect of AIST stronger for organisations that face more competitive pressure.	Supported

7.1. Contribution to research

This study makes several theoretical contributions. First, to the best of our knowledge, this study is among the first to introduce and explore the concept of organisational continuous improvement in information security management. Despite the importance of CIISM, there is no theoretically driven investigation into this concept. This study fills the literature gap. We call for more studies on organisational CIISM. A better understanding will enrich the literature and help organisations better prepare for evolving cyber-attacks. Moreover, this study adds to the literature on ISM by taking a dynamic capabilities perspective to explore how ACAP impacts CIISM. ACAP provides an underlining ongoing process wherein organisations employ the latest obtained knowledge to bolster their resources and processes while steadily enhancing their ISMS. The finding that ACAP affects CIISM is in line with earlier findings that introduce learning as the prerequisite of continuous improvement (e.g., GARVIN, 1993). This study improves our understanding of continuous improvement in an ever-changing competitive context such as cybersecurity, where organisations repeatedly encounter new challenges and threats and highlights the important role of ACAP in enhancing continuous improvement activities in such every-changing contexts.

Second, this study conceptualises and empirically tests the mediation role of AIST as a form of strategic flexibility in the context of information security. This advancement is particularly important in the field of information security as it helps organisations consider the necessity of the ability to adapt to the latest security threats and know this adaptability can mediate the effect of their learning capability on organisational outcomes. In other words, the mediation results indicate that flexibility is not a plus capability but a requirement for organisations that want to channel the effect of ACAP to continuous improvement. In addition, previous studies have indicated that when the effect of learning is not obvious, it might be unclear the need to learn, what to learn, and where to apply this learning (NI & SUN, 2009). Introducing AIST as a mediator in the context of information security allows organisations to align the learned knowledge more towards continuous enhancement in ISMS. Further, presenting AIST as an organisation's strategic flexibility in the context of information security fill up the existing literature gap in terms of a form of dynamic capability that can handle security threats by using both existing skills and renewed competencies concurrently.

Third, the results of the study lend empirical support to the idea that competitive pressure can strengthen ACAP's effect on adaptability. Indeed, the mediating effect of AIST is stronger when there is higher competitive pressure. This key finding provides evidence that the interaction of organisational capabilities and the external environment can be beneficial and facilitate flexibility in anticipating new information security threats and risks. Competitive pressure plays an important contingency factor that could alter an organisation's strategic response regarding the importance of information security. It may lead to being more proactive in managing security practices. This finding is the first step towards investigating the relationship between environmental factors, dynamic organisational capabilities, and continuous improvement. Results also fill the literature gap in organisational security management, as few studies have looked into how environmental factors may change an organisation's security management, although there are abundant anecdotal discussions.

7.2. Implications for practice

Information security management is a complex dynamic problem. The increasing dependence on information systems has made organisations a more attractive target of constantly evolving security threats. If organisations fail to improve their security management continuously, their approaches and practices may become outdated. Consequently, they become vulnerable. This study calls for organisations to be mindful of their continuous improvement in ISM. Organising workshops and training sessions can help increase firm-wide awareness about the necessity of CIISM and effective methods to achieve it.

The results of this study have several applicable implications for managers and policymakers. First, it is critical for executives and managers to uncover and develop organisational capabilities that facilitate the

continuous improvement process. Our findings present evidence on how ACAP could lead to such outcomes. Investing resources in building up ACAP may equip organisations with the necessary skills to continually improve their ISM practices. Competition and cyber espionage in terms of theft of confidential information and intellectual property have been some of the main motives behind cyberattacks (BOSHELL, 2021; MARK, 2020). This result is especially important for organisations active in hypercompetitive environments. As our results have shown, the effect of ACAP is amplified in such an environment.

Furthermore, the results of this study may motivate organisations to recognise the importance of developing strategic flexibility in ISM. Such flexibility not only helps organisations adapt to the latest security threats but also enhances organisational outcomes in the long term. Providing facts and real-world examples to organisations along with guidelines on recognising and gathering relevant information, the latest techniques to understand this information, and how to integrate and apply it could be an effective method to encourage devoting resources and effort to put AIST into practice.

7.3. Limitations and future research

Last but not least, we acknowledge several limitations of the study, which offer opportunities for future research. First, the data came from a single source, i.e., managers with information security knowledge assessed both dependent and independent variables in one survey, which makes the findings subjective in nature. To address the concern of common method bias (CMB), we have taken several steps. Even though our data analysis results suggest that CMB is not a serious concern, future empirical studies can improve upon this limitation by using data collected from multiple sources. In particular, future studies are encouraged to employ approaches that can objectively measure dependent and independent variables.

Second, the study was conducted in the United States, which may raise the possibility of regional bias. Moreover, approximately 73.8% of our participants are Caucasian. Although the proportion of Caucasian managers is very similar to that of the United States workforce (i.e., 72%),[3] scholars should be cautious when interpreting results for organisations with operations in countries with a different culture or ones managed by non-Caucasian individuals. Moreover, among the industries from which we recruited study participants, education (6.9%), finance (11.5%), retail (10.8%), and healthcare (10%) are the most common. In addition, over 60% of the managers came from other industries, including construction and insurance. It is important to note that certain characteristics unique to a particular industry may affect the generalisability of our results. Although the research hypotheses have received strong empirical support, we believe there are viable future opportunities to replicate and extend the current study in the context of other countries and economies.

Lastly, as there is some unexplained variance in our dependent variable, CIISM (according to the coefficient of determination as suggested by our data analysis results), future studies should consider examining the effects of other unidentified organisational or environmental factors (e.g., organisational culture, public policies, etc.) that may provide additional insights into this important phenomenon. Besides, since both CIISM and AIST are new concepts in the information security literature, plenty of research lines drawing upon other theoretical perspectives are warranted. In addition, future empirical studies using mixed methods combining qualitative and archival data may also offer some new perspectives.

7.4. Conclusion

Grounded in the tenets of dynamic capabilities theory, this study explores drivers of continuous improvement of information security systems from an organisational learning perspective. This study shows that absorptive capacity affects continuous improvement of information security management systems directly and indirectly through an organisation's adaptability to information security threats. Moreover, competitive pressure enhances the mediating effect of an organisation's adaptability to information security threats. These findings offer significant contributions to the information security literature and have significant implications for executives, managers, and policymakers.

Notes

1. https://www.iso.org/isoiec-27001-information-security.html
2. https://www.firemon.com/continuous-improvement-key-network-security-management/
3. https://www.zippia.com/manager-jobs/demographics/
   https://datausa.io/profile/soc/management-occupations#demographics

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Appendix

Table A1. Instrument Items.

Construct	Items	Reference(s)
Dependent Variable		
Continuous Improvement in InfoSec Management	• CIISM1: In my organisation, we actively seek out areas for continuous improvement of our information security management. • CIISM2: In my organisation, we continuously revise information security management processes. • CIISM3: In my organisation, we are constantly on the lookout to improve our information security management effort.	(De JONG & De Ruyter, 2004)
Independent Variable		
Absorptive Capacity	• AC1: We are successful in learning new things within our organisation. • AC2: My organisation is effective in developing new knowledge or insights that have the potential to influence our business. • AC3: My organisation is able to identify and acquire internal (e.g., within the organisation) and external (e.g., market) knowledge. • AC4: My organisation has effective routines to identify, value, and import new information and knowledge. • AC5: My organisation has adequate routines to analyse the information and knowledge obtained. • AC6: My organisation has adequate routines to assimilate new information and knowledge. • AC7: My organisation can successfully integrate our existing knowledge with the new information and knowledge acquired. • AC8: My organisation is effective in transforming existing information into new knowledge. • AC9: My organisation can successfully exploit internal and external information and knowledge into concrete applications. • AC10: My organisation is effective in utilising knowledge in new services.	(PAVLOU & El Sawy, 2006)
Mediator		
Adaptability to InfoSec Threats	• AIST1: When we feel that one effort of managing information security risk is not working, we can easily change to another. • AIST2: In my organisation, we try to understand how one information security threat differs from another in risk management expectations. • AIST3: In my organisation, it is easy to modify our information security management approach if the situation calls for it.	(SPIRO & WEITZ, 1990; De JONG & De Ruyter, 2004)
Moderator		
Competitive Pressure	• CP1: My organisation has experienced competitive pressure to implement innovative information security measures. • CP2: My organisation will experience a competitive disadvantage if innovative information security methods are not adopted. • CP3: Competition is a factor in our decision to adopt innovative security measures. • CP4: Our industry is pressuring us to adopt innovative security measures.	(GRANDON & PEARSON, 2004; LIN, 2006)

Table A2. Factor Loadings and Cross-Loadings.

	CIISM	Absorptive Capacity	AIST	Competitive Pressure
CIISM1	**0.86**	0.60	0.65	0.58
CIISM2	**0.89**	0.62	0.67	0.60
CIISM3	**0.86**	0.59	0.65	0.57
AC1	0.53	**0.77**	0.57	0.35
AC2	0.51	**0.75**	0.55	0.33
AC3	0.50	**0.73**	0.54	0.33
AC4	0.53	**0.76**	0.56	0.34
AC5	0.54	**0.78**	0.58	0.35
AC6	0.54	**0.79**	0.58	0.35
AC7	0.55	**0.80**	0.59	0.36
AC8	0.58	**0.84**	0.62	0.38
AC 9	0.56	**0.81**	0.60	0.37
AC10	0.61	**0.88**	0.65	0.39
AIST1	0.58	0.57	**0.77**	0.47
AIST2	0.60	0.59	**0.79**	0.48
AIST3	0.56	0.55	**0.74**	0.45
CP1	0.54	0.36	0.49	**0.80**
CP2	0.53	0.36	0.48	**0.79**
CP3	0.52	0.35	0.47	**0.77**
CP4	0.54	0.36	0.49	**0.81**

Post-Hoc Analysis

We conducted additional tests to find out whether competitive pressure has a direct influence on absorptive capacity and adaptability to information security threats in addition to its moderating influence. Results indicate that competitive pressure has a significant positive effect on both AIST ($\beta = 0.33$, $p < .001$) and ACAP ($\beta = 0.41$, $p < .001$). However, since our focus and proposed framework did not concern these direct effects, we did not hypothesise them.

Common Method Bias Assessment

As the data were collected from a single source at a single point of time, common method bias could be a concern (PODSAKOFF et al., 2003). Following PODSAKOFF et al. (2003), we accounted for common method bias by employing procedural and statistical remedies (see, Table 9). Procedurally, we used pre-validated items; we had counterbalanced the order of questions and ensured respondent anonymity, and we also avoided using neutral or reverse-coded questions. Statistically, following LINDELL and WHITNEY (2001) and PODSAKOFF et al. (2003), we

added a marker variable and found that all significant correlations stayed significant, indicating that CMV is not a serious concern to our dataset (LINDELL & WHITNEY, 2001). Moreover, we conducted a full collinearity assessment (KOCK, 2015; KOCK & LYNN, 2012), which has also been used in previous studies in information systems literature (YAZDANMEHR et al., 2020). The results of the assessment are summarised in Table A3. The VIF values of all constructs were well below the cut-off of 10 recommended by previous research (NETER et al., 1989, p. 409), indicating that CMB is not a major concern..

Table A3. Procedural and Statistical Remedies to Prevent and Test Common Method Bias.

Procedural Remedies (PODSAKOFF et al., 2003; MACKENZIE et al., 2011)

- We used pre-validated reliable items from the previous well-established studies.
- We protected participants' anonymity by communicating that their responses would be confidential and that there were no right or wrong answers.
  We randomised the orders of the questions within each survey block as well as the survey blocks.

Statistical Remedies:

Partialling out a "marker" variable or correlational marker approach (LINDELL & WHITNEY, 2001; PODSAKOFF et al., 2003)

- We did not include any absolute unrelated items for the purpose of testing with a marker variable. However, we identified a potential marker variable that is theoretically unrelated to any factor in the model. We used the item, "Overall, I found organizational authentication systems to be usable.", as a marker variable since it is theoretically unrelated to other constructs used in our study, was not significantly correlated with other measurement items in our study, and was measured in the same fashion (i.e., Likert Scale) as other measurements in our study. We added a marker variable and linked it to all indicators of the other latent factors. We then computed the latent variables and ran the regression analysis. The results showed that all significant correlations stayed significant (LINDELL & WHITNEY, 2001). Taken together, these results indicated that CMB is not a major issue in our data.

Full Collinearity Assessment Approach (KOCK & LYNN, 2012; KOCK, 2015)

- We conducted the full collinearity test. The results showed that VIFs are all less than 2.2, which are well below the cut-off of 10 as recommended (Neter et al. (NETER et al., 1989), p. 409), suggesting that CMB is not a major concern.

VIF Results for Full Collinearity Assessment

	AIST	ACAP	Competitive Pressure	CIISM
AIST		1.92	2.15	2.14
ACAP	1.69		2.07	1.85
Competitive Pressure	1.54	1.63		1.41
CIISM	2.14	2.09	2.04	

Table A4. VIF Values for Interaction Effects.

	DV: AIST	
Predictors		
	ACAP	VIF = 1.31
	Competitive Pressure	VIF = 1.24
	ACAP x Competitive Pressure	VIF = 1.18 *